THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 7 & 8: **Proportional Reasoning-1, Fractions in Disguise** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 4 : 5	Divide both by HCF 7
2	**(d)** 20	Product of extremes = product of means: $3x = 60$
3	**(c)** 24	Product of extremes = product of means: $7x = 168$
4	**(a)** ₹ 360	A gets 3/7 of 840
5	**(b)** 0.75 kg	Unit conversion
6	**(b)** 62.5%	Multiply by 100
7	**(c)** 210	35/100 × 600
8	**(b)** 60	7.5/100 × 800
9	**(d)** 1640	2000 × 82/100
10	**(b)** 25%	Profit or loss on CP

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	1 unit on the map is 50000 of the same units on the ground (2).	2
12	6 cups (2).	2
13	$\frac{1}{8}$ (2).	2
14	SI ₹ 1000, amount ₹ 6000 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$6 \times 200000 = 12$ lakh cm (1); $= 12$ km (2).	1+2
16	$A = P(1 + r/100)^n = 8000 \times (1 + 5/100)^3 = ₹9,261.00$ (2); CI $= ₹1,261.00$ (1).	2+1
17	$7/20 \times 5 = 1.75$ kg (3).	3
	OR: 14 km per L (1); 210 km (2).	1+2
18	$A = P(1 + r/100)^n = 12000 \times (1 + 6/100)^2 = ₹13,483.20$ (2); CI $= ₹1,483.20$ (1).	2+1
	OR: SI $= ₹1350$ (2); amount $= ₹8850$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 3 hours (3). (ii) Inverse proportion (2).	3+2
	OR: (i) 21 L (2). (ii) 30 L (1). (iii) 30% (2).	2+1+2
20	(i) Percentage of the value of goods or services added to it (2). (ii) ₹ 1890 (3).	2+3
	OR: (i) ₹ 135 (1). (ii) 15% (2). (iii) ₹ 688.50 (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 7 & 8: **Proportional Reasoning-1, Fractions in Disguise** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** division	Ratio $a : b = a/b$
2	**(a)** 9 : 12	Multiply both by 3
3	**(d)** 1 : 8	15 : 120
4	**(c)** ₹ 192	₹ 24 each
5	**(b)** becomes half	Product constant
6	**(d)** ₹ 1800	SP = MP − discount
7	**(c)** ₹ 1440	$SI = \frac{PRT}{100}$
8	**(b)** 45%	× 100
9	**(b)** less than the original	0.99 of original
10	**(d)** 225	45 × 5

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Reduce to simplest form (1); cross-multiply (1).	1+1
12	$120 : 40 = 3 : 1$ (2).	2
13	Interest calculated on the principal and on the interest earned earlier (2).	2
14	42000 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	21 and 15 (3).	3
16	SI = ₹1800 (2); amount = ₹10800 (1).	2+1
17	8 m, 12 m, 16 m (3).	3
	OR: 240 km (3).	3
18	SI = ₹840 (2); amount = ₹4340 (1).	2+1
	OR: SI = ₹480 (2); amount = ₹2880 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Ratio $6 : 4 : 3$; parts 72, 48 and 36 (3). (ii) The first (2).	3+2
	OR: (i) $1 : 2$ and $4 : 5$ (3). (ii) $4 : 5$ is greater since $0.8 > 0.5$ (2).	3+2
20	(i) $P \times 1.1664 = 5832$, $P = 5000$ (3). (ii) Interest added to principal each period (2).	3+2
	OR: (i) 24 girls, 6 with bicycles (3). (ii) 15% (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET C

Chapters 7 & 8: **Proportional Reasoning-1, Fractions in Disguise** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2 : 5	Divide both by HCF 18
2	**(a)** 45	Product of extremes = product of means: $4x = 180$
3	**(b)** ₹ 200	A gets 2/5 of 500
4	**(d)** ₹ 300	A gets 5/11 of 660
5	**(a)** 4200 mL	Unit conversion
6	**(c)** 36%	Multiply by 100
7	**(b)** 60	12.5/100 × 480
8	**(a)** 1000	800 × 125/100
9	**(d)** 540	450 × 120/100
10	**(d)** 20%	Profit or loss on CP

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$24 : 6 = 4 : 1$ (2).	2
12	15 years (2).	2
13	$360/1.2 = 300$ (2).	2
14	37.5% (1); 0.375 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$320 : 160 = 2 : 1$ (3).	3
16	New price 1.2; spending same so quantity $1/1.2 = 83.3\%$ (2); falls by 16.67% (1).	2+1
17	Ages 25 and 35 (2); after 5 years 30 and 40 (1).	2+1
	OR: Total work 180 man-days (1.5); $180/20 = 9$ days (1.5).	1½+1½
18	72% (2); $\frac{18}{25}$ (1).	2+1
	OR: 360 play (1.5); 840 do not (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) ₹ 15 (1). (ii) ₹ 210 (2). (iii) 25 pens (2).	1+2+2
	OR: (i) 1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm (2). (ii) 4250 m and 4500 cm (3).	2+3
20	(i) $1.1 \times 1.1 = 1.21$ (3). (ii) $1.1 \times 0.9 = 0.99$, a 1% decrease (2).	3+2
	OR: (i) ₹ 18000 (2). (ii) ₹ 68000 (1). (iii) ₹ 1888.89 (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

Chapters 7 & 8: **Proportional Reasoning-1, Fractions in Disguise** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 5 : 2	100 cm : 40 cm
2	**(d)** ₹ 150	₹ 15 per pen
3	**(a)** 8 : 12 : 15	Make B equal
4	**(c)** 1 : 8	250 g : 2000 g
5	**(d)** ₹ 135	₹ 7.5 each
6	**(b)** ₹ 800	$SI = \frac{PRT}{100}$
7	**(d)** ₹ 3600	$SI = \frac{PRT}{100}$
8	**(d)** interest is also earned on earlier interest	Compounding
9	**(a)** goods and services	Goods and Services Tax
10	**(c)** ₹ 1210	1000 × 1.1 × 1.1

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$45 : 1440 = 1 : 32$ (2).	2
12	When one quantity increases, the other increases in the same ratio (1); cost and quantity (1).	1+1
13	GST ₹ 90; total ₹ 590 (2).	2
14	₹ 60 (1); 10% (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Speed 60 km/h (1.5); 4.5 hours (1.5).	1½+1½
16	Loss ₹ 500 (2); SP ₹ 3500 (1).	2+1
17	Milk 35 L, water 5 L (1); new water $= 15$ L (1); add 10 L (1).	1+1+1
	OR: Sides 2 : 3 (1.5); areas 4 : 9 (1.5).	1½+1½
18	Salary ₹ 30000 (2); annual savings ₹ 54000 (1).	2+1
	OR: $R = \frac{100 \times 1920}{8000 \times 3}$ (2); 8% (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Incomes $5x$, $4x$, expenses $4y$, $3y$; $5x - 4y = 1000$, $4x - 3y = 1000$ give $x = y = 1000$, incomes ₹ 5000 and ₹ 4000 (3). (ii) Expenses ₹ 4000 and ₹ 3000, savings ₹ 1000 each (2).	3+2
	OR: (i) $49 : 196 = 1 : 4$ (3). (ii) Sides $1 : 2$, so area ratio is the square of the side ratio (2).	3+2
20	(i) 37.5% and 62.5% (2). (ii) 28 L (3).	2+3
	OR: (i) 12000 then 10800 (3). (ii) 28% (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.